INCH-POUND
MIL-DTL-24643/75A
27 November 2023
SUPERSEDING
MIL-DTL-24643/75
13 December 2011

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSLCFS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

These are low-capacitance, foil-shielded twisted pairs intended for use with serial interfaces defined by ANSI/TIA-232, TIA-530, TIA-422, and TIA-485, and are intended to connect computers and peripheral devices from a 25- or 50-pin grounded serial interface.

Construction (Non-Watertight)

First	-	Copper conductor tin-coated, 24 AWG, ASTM B8, class B stranding, 7 strands.
Second	-	Thermoset insulation. Wall thickness as necessary for the completed cable to meet the specified electrical requirements. Maximum diameter shall not exceed 0.055 inch. Solid color code to be applied using method 3; see table II for colors required.
Third	-	Two conductors twisted together to form a pair with a lay not exceeding 1.50 inches. The twisting of the individual pairs shall be such that the completed cable meets the electrical requirements specified herein.
Fourth	-	The specified number of pairs and singles (see table I) shall be cabled together with a lay (or lays) as required for the completed cable to meet the electrical requirements specified herein. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers are not permitted in these cables, but the cabled cores shall still form a firm, well-rounded assembly. Cabling layup shall be as follows: a. LSLCFS-25: Pairs 1 to 3 as the core layer and pairs 4 to 12 in sequence as the outer layer. The single conductor shall be placed in one outer valley of the core layer. b. LSLCFS-50: Pairs 1 to 3 as the core layer, pairs 4 to 10 in sequence as the second layer, and pairs 11 to 25 in sequence as the outer layer.
Fifth	-	Binder tape applied with overlap.
Sixth	-	Drain wire, copper conductor, tin-coated, stranded, 24 AWG, ASTM B8, class B stranding, 7 strands.

Distribution Statement A. Approved for public release: distribution is unlimited.

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Seventh - Aluminum foil/polyester tape. Metallic foil side facing in and in contact with the drain wire. Tape may be longitudinally applied with 10 percent minimum overlap or helically applied with 25 percent minimum overlap.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/75	Type and size	Number of pairs	Number of singles	Overall diameter (inch)	
				min.	max.
-01UO	LSLCFS-25	12	1	0.450	0.530
-02UO	LSLCFS-50	25	0	0.490	0.570

TABLE II. Identification color code.

Component	Conductor 1	Conductor 2	Component	Conductor 1	Conductor 2
Single	White	- - -	Pair 13	Black	Green
Pair 1	Blue	White	Pair 14	Black	Brown
Pair 2	Orange	White	Pair 15	Black	Gray
Pair 3	Green	White	Pair 16	Yellow	Blue
Pair 4	Brown	White	Pair 17	Yellow	Orange
Pair 5	Gray	White	Pair 18	Yellow	Green
Pair 6	Red	Blue	Pair 19	Yellow	Brown
Pair 7	Red	Orange	Pair 20	Yellow	Gray
Pair 8	Red	Green	Pair 21	Purple	Blue
Pair 9	Red	Brown	Pair 22	Purple	Orange
Pair 10	Red	Gray	Pair 23	Purple	Green
Pair 11	Black	Blue	Pair 24	Purple	Brown
Pair 12	Black	Orange	Pair 25	Purple	Gray

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) 28.50

Voltage withstand (volts, root mean square, min.)

Conductor to conductor 2,000

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Conductor to overall shield	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 meters)	52
Between conductor and shield at 1 kHz (pF/meter, max.)	80
Character impedance, at 1 megahertz (ohms)	100±15
Group B:	
Cold bending (cable)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Shield conformance to material and application	No failure
Permanence of printing (jacket) (cycles, min.)	125

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; Groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Insulation	45
Toxicity index (max.)	
Jacket	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSLCFS (all sizes)	1,000

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2015-032)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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